

Trainings ▾

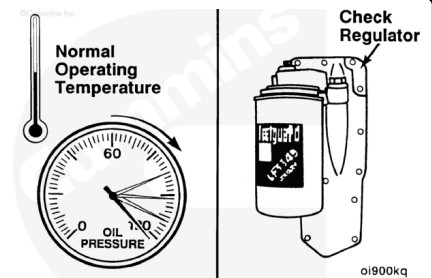
General Information

Lubricating Oil Pressure Regulating Valve

The engine will have high oil pressure at normal operating temperature if the regulator sticks in the closed position. Check the regulator for freedom of movement.

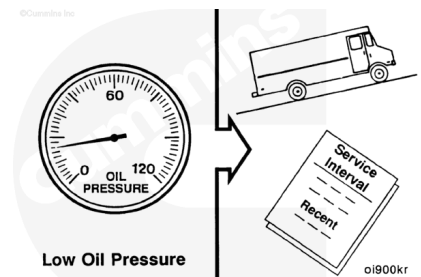
Refer to Procedure 007-029 in Section 7.

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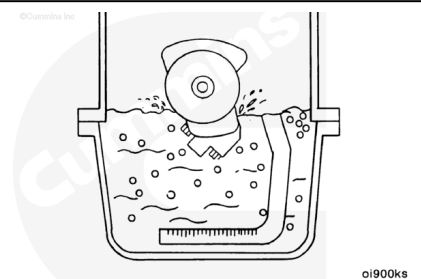
Low-Lubricating Oil Pressure

Low-lubricating oil pressure (or no oil pressure) can be caused by several lubrication system related malfunctions. To begin the investigation, determine the engine operating conditions when the low-pressure was first observed; i.e., following a service interval, at idle **only**, or while operating on a steep grade.

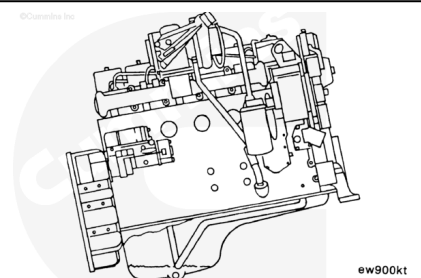


Lubricating Oil Level

High oil level can cause low oil pressure. If the oil level is high enough for the connecting rods to dip into during operation, the oil can become aerated, resulting in low oil pressure.



Low-lubricating oil level will **not** normally appear as low oil pressure. Typically, it will appear as an intermittent loss of oil pressure when rounding a corner or operating on a steep grade. This condition exists when the oil level is extremely low and the suction tube can **not** pick up oil during all modes of operation.



Lubricating Oil Coolers

The engines use full-flow, plate-type oil coolers. The oil flows through a cast passage in the cooler cover and through the element, where it is cooled by engine coolant flowing past the plates of the element.

Depending on the engine configuration, the number of lubricating oil cooler element plates will vary.

For front gear train engines:

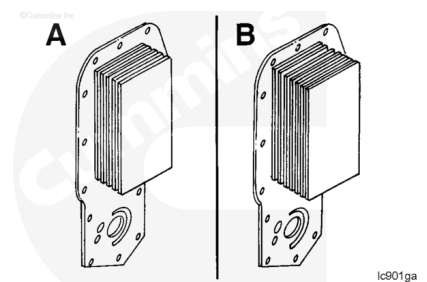
- Four-cylinder engines use an element with five plates (A)
- Six-cylinder engines use an element with seven plates (B).

For rear gear train engines:

- Four-cylinder engines use an element with seven plates (**Not Shown**).

Because of the differences in plate restriction and oil pump capacities, the oil cooler components are **not** interchangeable between the four-cylinder and six-cylinder engines or between the front and rear gear train engines. The use of incorrect components can cause high or low oil temperature, varnish, and sludge buildup.

Note : Prior to October 10, 1986, six-cylinder engines were assembled with nine-plate oil cooler elements.

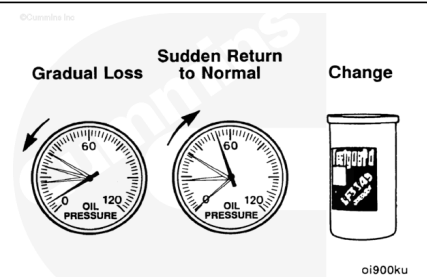


Lubricating Oil Filter Bypass Valve

The lubricating oil cooler cover contains a bypass valve that will let the lubricating oil flow bypass a plugged filter. The valve is designed to open when the pressure drop across the filter becomes too high, as with a plugged filter, and lets the lubricating oil continue on through the engine. When a filter becomes plugged, an oil pressure decrease of 60 kPa [9 psi] or less from the normal operating pressure can be observed on the vehicle lubricating oil pressure gauge. This allows unfiltered oil into the engine. This condition should be avoided by changing the filter at each oil drain interval, as described in the Operation and Maintenance Manual.

⚠ CAUTION ⚠

Using a filter for a six-cylinder engine on a four-cylinder engine does not allow extended change interval. Never use a filter for a four-cylinder engine on a six-cylinder engine



because of the reduced material holding capability and increased chance of plugging. Failure to do so can cause engine damage.

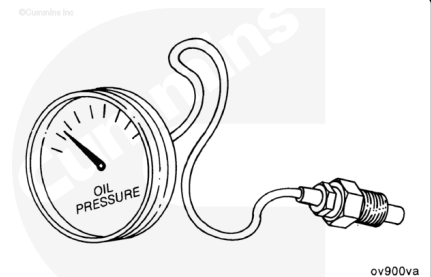
Lubricating Oil Filter

After the oil is cooled, it flows through the full-flow oil filter.

A plugged filter will cause a gradual loss of oil pressure by approximately 69 kPa [10 psi]. The pressure will return to normal when the filter bypass valve opens. If **not** corrected, this will result in severe engine wear, as the engine is running on unfiltered oil when the bypass valve is open.

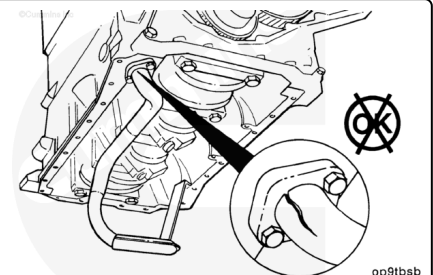
Lubricating Oil Gauge

Check the lubricating oil gauge and sending unit to make sure they are operating correctly by verifying the pressure with a manual gauge connected to the main oil rifle.



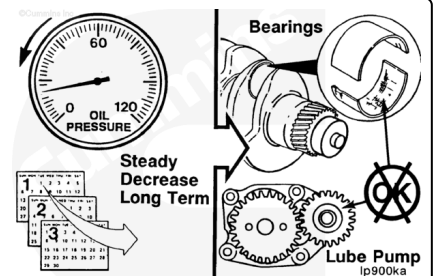
Lubricating Oil Suction Tube

A loose suction tube, damaged gasket, or crack in the suction tube can cause a temporary loss of prime for the oil pump. The engine will have low pressure or no oil pressure at starting, followed by normal oil pressure.



Bearings and Lubricating Oil Pump

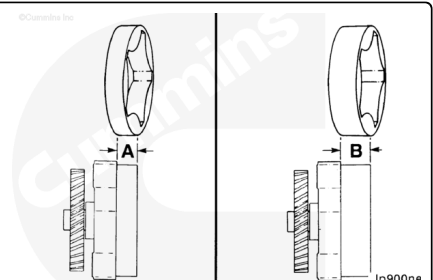
A steady decrease in oil pressure over a long period can be an indication of worn bearings or excessive oil pump wear.



Incorrect Lubricating Oil Pump

The capacity of the six-cylinder oil pump is greater than the four-cylinder. If low or high pressure occurs after changing the pump, verify that the correct pump was used. Refer to Procedure 007-031 in Section 7.

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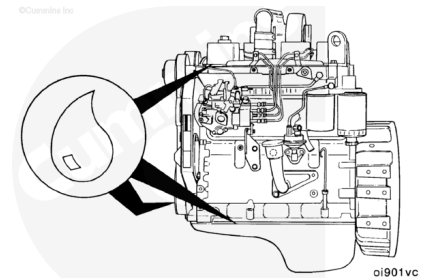


The flow capacity of a lubricating oil pump is typically controlled by the gerotor thickness, shown by A and B in the illustration.

Note : Earlier model B Series engines used lower capacity pumps. Check to be sure the correct model pump is used.

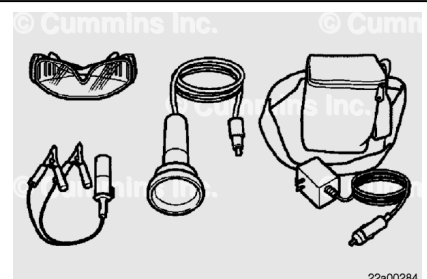
Lubricating Oil Leaks

Various gaskets, seals, and plugs are used to contain the lubricating oil. Most leaks can be identified during routine inspection of the engine and vehicle.



A black light kit, Part Number 3163338, can be very helpful when locating the source of an oil leak.

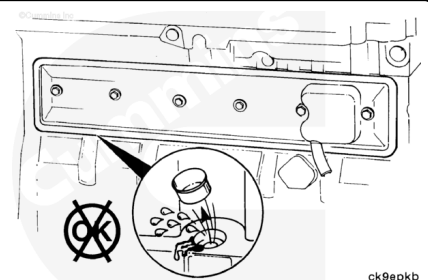
The fluorescent tracer, Part Number 3376891, will be highlighted by the black light to help identify the source of the leak.



A blown expansion plug can allow a relatively large quantity of lubricating oil to escape, resulting in a sudden drop in the lubricating oil pressure.

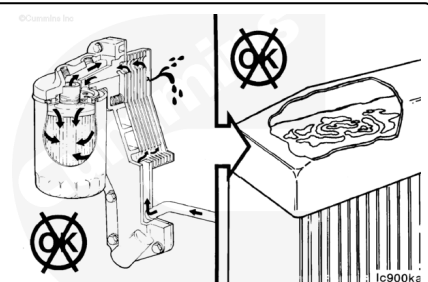
For front gear train engines, when checking for such a leak, be sure to check the expansion plug behind the tappet cover. Lubricating oil blowing out the breather is a good sign of a blown expansion plug. This is usually more obvious on a four-cylinder engine. Refer to Procedure 004-017 in Section 4 to remove the tappet cover.

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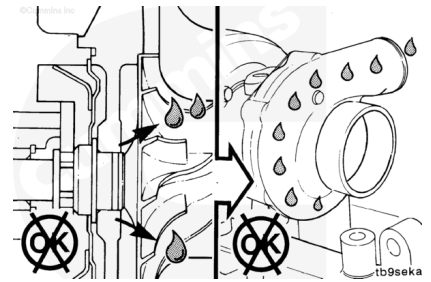
If the lubricating oil cooler element ruptures, the lubricating oil pressure will force lubricating oil into the coolant system.

Refer to Procedure 007-003 in Section 7
(/qs3/pubsys2/xml/en/procedures/40/40-007-003-tr.html) to inspect the lubricating oil cooler.



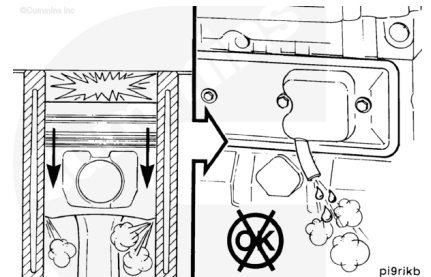
If equipped, worn or damaged seals in the turbocharger can also allow lubricating oil to leak into the air crossover pipe and be burned in the engine.

The condition can be verified by removing the air crossover tube or charge air cooler tubing and looking for oil.



Inadequate sealing of the piston rings will result in lubricating oil being blown out the breather tube and consumed by the engine. Refer to Procedure 014-010 in Section 14.

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If equipped, lubricating oil can also be lost through a worn or malfunctioning air compressor. Look for carbon buildup in the air line from the compressor to the air tank. Also, a failed air compressor head or head gasket can allow oil to leak into the coolant or coolant to leak into the oil during hot shutdown.

Refer to Procedure 012-003 in Section 12.

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